SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

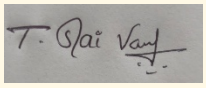
ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA0310	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	T. Sai varshini	**Reg. No.:**	192472403
Section / Batch:	Slot B	**Date:**	30-10-2026

Code of Conduct

I T.SAI VARSHINI / 192472403 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

1. Design a concept map that evaluates the relationship between frequency reuse, interference, and network capacity.
   Assess how each factor influences cellular system efficiency.
   Prioritize the concepts based on their impact.
   Justify the relationships shown in the concept map.

2. Construct a concept map that evaluates the interdependence among cellular network architecture, protocol stack, and mobility management.
   Assess how these components collectively improve communication reliability.
   Identify the strongest conceptual relationships.
   Justify your evaluation using appropriate linkages.

3. Develop a concept map that evaluates resource management strategies in cellular networks.
   Relate channel allocation, power control, interference management, frequency reuse, and handoff management.
   Assess which strategy contributes most to efficient network utilization.
   Support your evaluation through concept relationships.

4. Design a concept map that evaluates the integration of all major concepts in mobile computing.
   Include architecture, protocol stack, mobility management, roaming, handoff, performance criteria, and cellular operations.
   Assess how these concepts collectively support an efficient mobile communication system.
   Justify the hierarchy and linking phrases used.

5. Construct a concept map that evaluates alternative approaches for improving cellular network performance.
   Relate mobility management, handoff optimization, interference reduction, protocol efficiency, and resource allocation.
   Recommend the most effective combination of techniques.
   Justify your recommendation through well-defined conceptual relationships.

Rubrics for Calculation

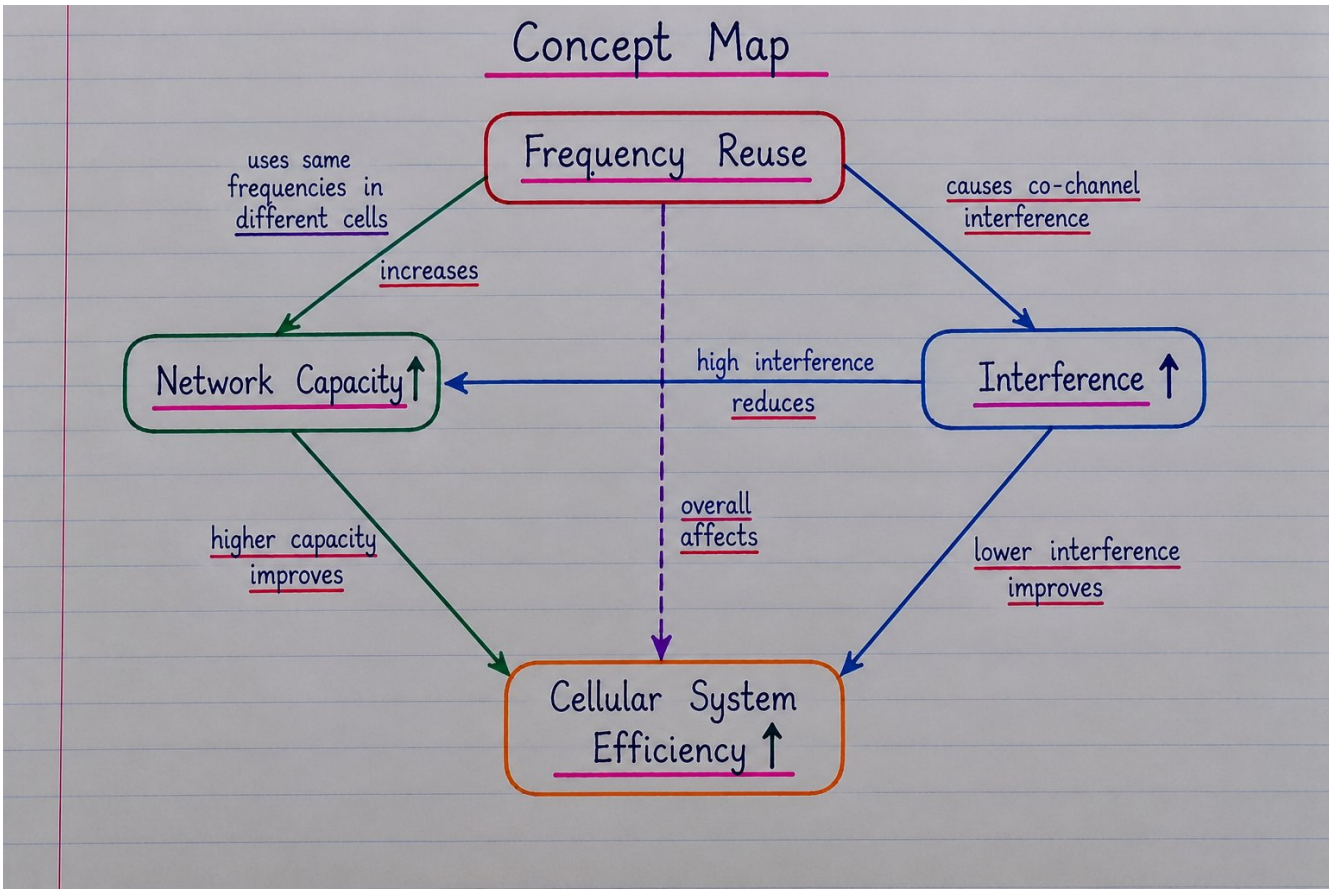
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question 1– Frequency Reuse, Interference and Network Capacity

Question:

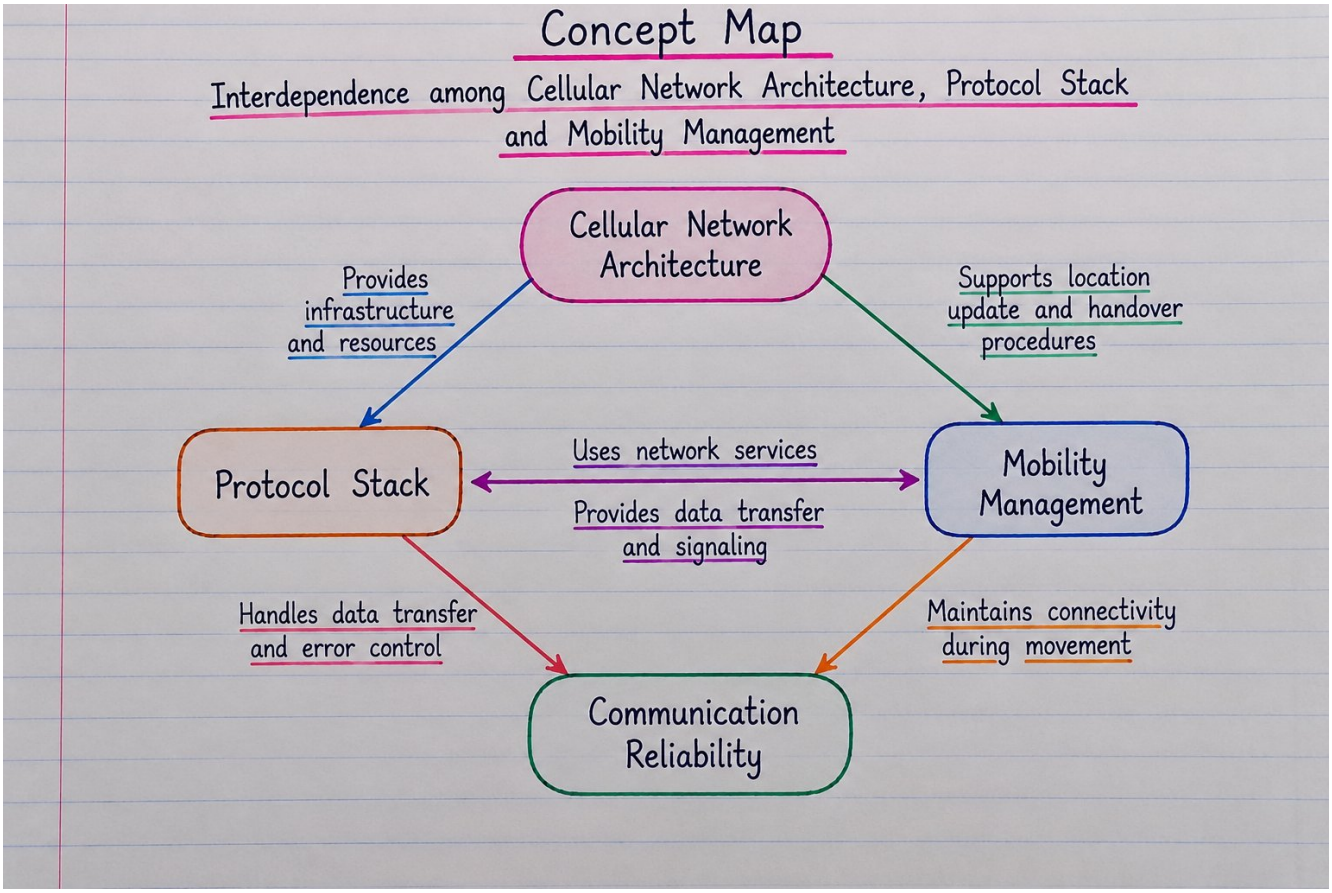
Design a concept map that evaluates the relationship between frequency reuse, interference, and network capacity. Assess how each factor influences cellular system efficiency. Prioritize the concepts based on their impact. Justify the relationships shown in the concept map.



Question 2– Cellular Network Architecture, Protocol Stack and Mobility Management

Question:

Construct a concept map that evaluates the interdependence among cellular network architecture, protocol stack, and mobility management. Assess how these components collectively improve communication reliability. Identify the strongest conceptual relationships. Justify your evaluation using appropriate linkages.

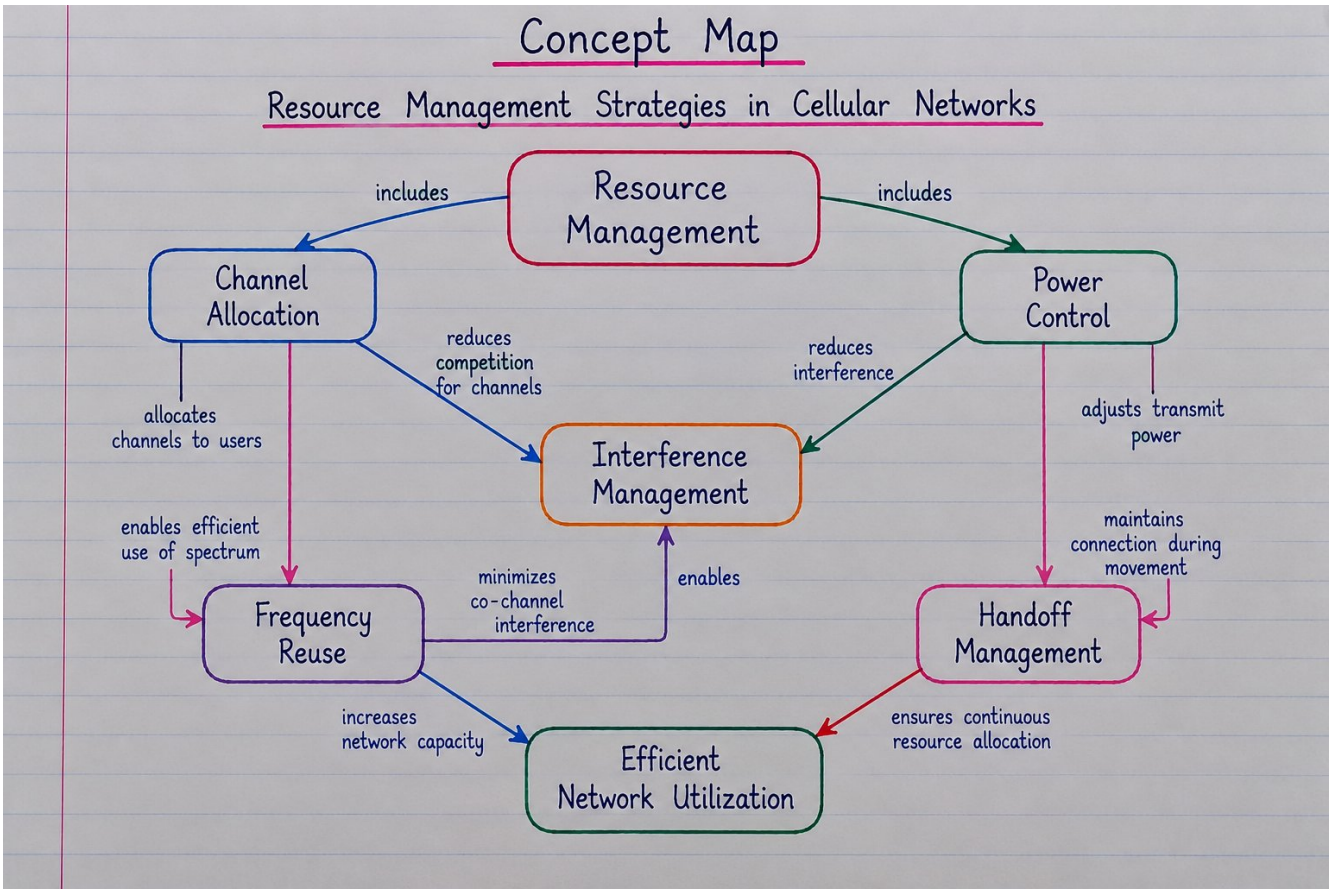


Question 3– Resource Management Strategies in Cellular Networks

Question:

Develop a concept map that evaluates resource management strategies in cellular networks. Relate channel allocation, power control, interference management, frequency reuse, and handoff management. Assess which strategy contributes most to efficient network utilization. Support your evaluation through concept relationships.

Concept Map

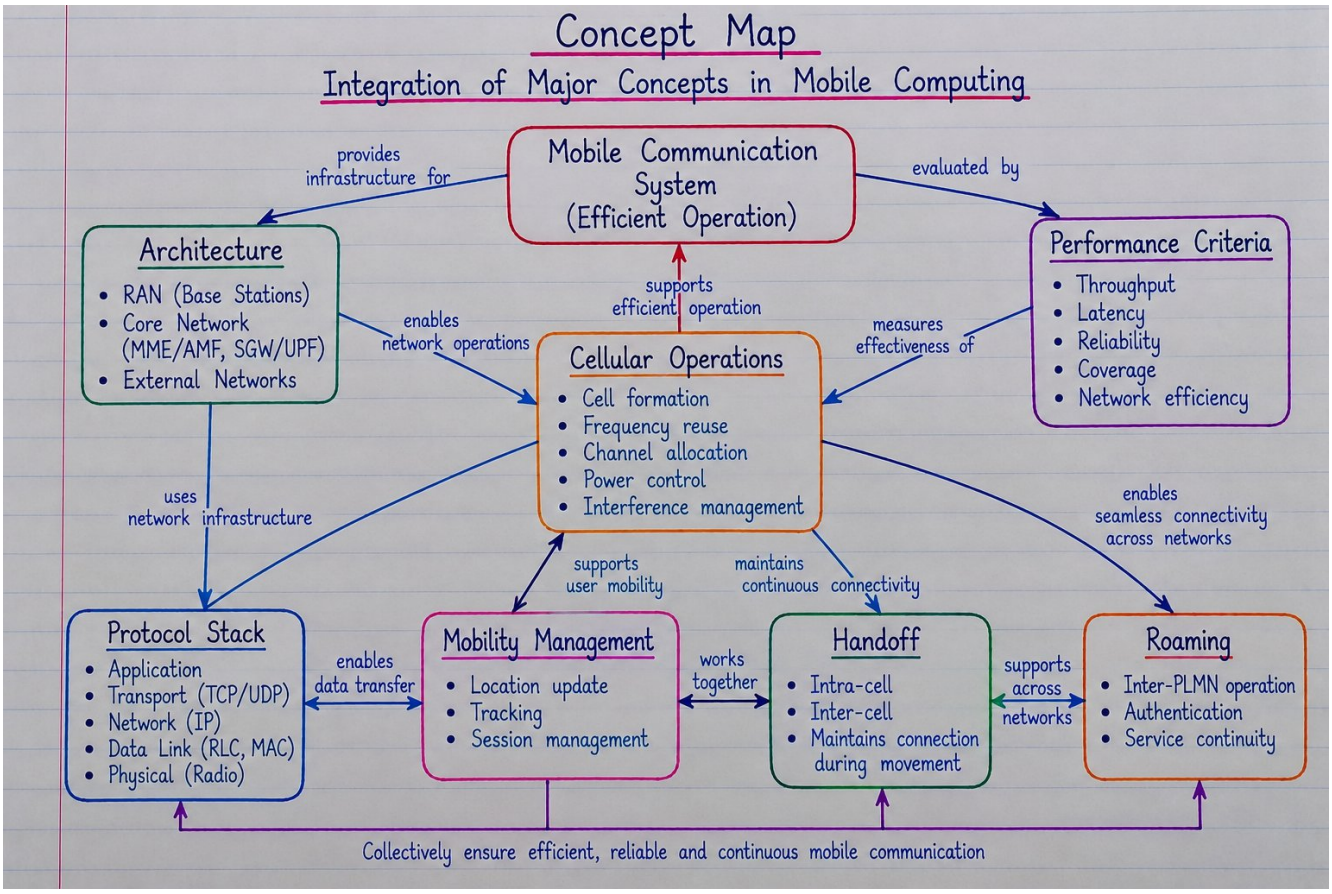


Question 4– Integration of Major Concepts in Mobile Computing

Question:

Design a concept map that evaluates the integration of all major concepts in mobile computing. Include architecture, protocol stack, mobility management, roaming, handoff, performance criteria, and cellular operations. Assess how these concepts collectively support an efficient mobile communication system. Justify the hierarchy and linking phrases used.

Concept Map



Question 5– Alternative Approaches for Improving Cellular Network Performance

Question:

Construct a concept map that evaluates alternative approaches for improving cellular network performance. Relate mobility management, handoff optimization, interference reduction, protocol efficiency, and resource allocation. Recommend the most effective combination of techniques. Justify your recommendation through well-defined conceptual relationships.

Concept Map

